

Muhammad Usama

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INTRODUCTION

AI Research Scientist specializing in 3D computer vision, shape reverse-engineering, and generative CAD. Backed by 3+ years of full-stack machine learning engineering and top-tier publications (AAAI, CVPR, ECCV).

EXPERIENCE

- **Intern Research Scientist** Los Altos, United States
Toyota Research Institute (TRI) *Jun 2026 – Present*
 - **Generative Reverse Engineering:** Architecting a novel vision-based framework for reverse-engineering automotive parts into highly editable CAD models that preserve design intent, with the goal of submission to CVPR 2027.
 - **Parametric CAD Generation:** Developing a NURBS-based generation framework for Toyota Motor Corporation (TMC) to translate non-editable geometry ("dumb solids") into fully parametric, boundary representation (BRep) CAD models for automotive car roofs.
 - **Deformed CAD Pipeline (POC):** Engineered an end-to-end Python pipeline using OpenCASCADE and Trimesh to bridge the gap between simulation-based mesh morphing and CAD updates. Implemented algorithms to map tessellated mesh triangles to BRep face IDs, compute displacements, and apply global affine transformations.
- **Student Research Assistant** Kaiserslautern, Germany
German Research Center (DFKI) *Nov 2023 – May 2026*
 - **Automated captioning pipeline :** Built an end-to-end pipeline to auto-caption **8.9M 3D assets** using vision-language models (VLMs); orchestrated multi-view renders and metadata extraction with Blender.
 - **Fine-tuning LLMs :** Researched and implemented **fine-tuning** strategies for large language models (LLMs), including Qwen-3B and Qwen-7B, using parameter-efficient methods such as **LoRA**.
 - **Long Context Fine-tuning :** Explored techniques for long-context fine-tuning of LLMs, evaluating approaches like **LongLoRA** and other context-extension strategies to enhance model performance on long-sequence tasks.
 - **Ethical filtering :** Designed an LLM-based ethical filtering layer with prompt engineering to detect, redact, and rewrite problematic captions; reduced ethical concerns by **80%** relative to the unfiltered output.
 - **Pose-guided image generation:** Gained in-depth knowledge of diffusion models through extensive research, evaluated existing methodologies and continuously analyzed ongoing advancements in pose-guided image generation.
 - **Visual Machine Learning:** Built backend **microservices** for a visual ML platform to create, train, and deploy models, supporting both the DFKI cluster and on-prem MindGarage CPU nodes.
- **Machine Learning Infrastructure** Redmond, United States
Retrocausal *Jul 2022 – Oct 2023*
 - **Software Packaging:** Led the software packaging for client deployment, tailoring systems to diverse factory environments. This played a key role in expanding our client base and securing **venture capital funding in late 2023**, marking a significant career milestone that reflects the global scale and recognition of my work.
 - **System Scalability:** Improved system scalability and efficiency by replacing continuous polling with **Redis-** and **SQS-**based messaging, enabling on-demand data loading and significantly reducing API calls for long-running and continuous tasks.
 - **Labelling Tool:** Built an efficient object labelling method for the system, enabling accurate labelling of 2D static and 3D objects in depth and RGB frames. Designed an intuitive UI, resulting in time savings and enhanced product appeal.
 - **On-Premises Architecture Design and Development:** At a major client's request, migrated the architecture from cloud to **on-premises** with minimal performance loss, replacing AWS with **Localstack** and alternatives to Firebase and other cloud services, and built a packaging script for seamless installation on client PCs.

- **Thin Web Client** : Developed a lightweight web client version to replace the bulky desktop application UI. Implemented **WebRTC** for seamless streaming and data channel communication.

- **Software Engineer**

Karachi, Pakistan

Aga Khan Development Network Digital Health Resource Centre (AKDN)

Jul 2020 – Mar 2022

- **Full-Stack Development**: Built a scalable healthcare system (records, scheduling, billing, inventory, payments) for the largest hospital network in Karachi and Nairobi using **.NET**, **NestJS**, and **Angular**.
- **Deployment Automation & Containerization**: Modernized deployment workflows by introducing **Docker**, **Docker Compose**, and **GitHub Actions**, reducing manual steps and production errors.

PUBLICATIONS

- **NURBGen**: **Muhammad Usama**, Mohammad Sadil Khan, Didier Stricker, and Muhammad Zeshan Afzal (2026). *NURBGen:High-Fidelity Text-to-CAD Generation through LLM-Driven NURBS Modeling*. **AAAI 2026** [\[Paper\]](#)
- **BRepClip**: **Muhammad Usama**, Didier Stricker, Mohammad Sadil Khan, Muhammad Zeshan (2026). *BRepCLIP: Contrastive Multimodal Pretraining on BRep Primitives for CAD Understanding* **Under Review** [\[Paper\]](#)
- **MARVEL-40M+**: Sankalp Sinha, Mohammad Sadil Khan, **Muhammad Usama**, Shino Sam , Didier Stricker , Sk Aziz Ali, and Muhammad Zeshan Afzal. (2025). *MARVEL-40M+:Multi-Level Visual Elaboration for High-Fidelity Text-to-3D Content Creation*. **CVPR 2025** [\[Paper\]](#)
- **DreamCAD**: Mohammad Sadil Khan, **Muhammad Usama**, Rolandos Alexandros Potamias, Didier Stricker, Muhammad Zeshan Afzal, Jiankang Deng, Ismail Elezi (2026). *DreamCAD:Scaling Multi-modal CAD Generation using Differentiable Parametric Surfaces*. **ECCV 2026** [\[Paper\]](#)
- **Physics-in-the-Loop**: Elias Berger, **Muhammad Usama**, Jan Mehlstaubl, Bernhard Saske , Kristin Paetzold-Byhain (2026). *Physics-in-the-Loop: A Hybrid Agentic Architecture for Validated CAD Engineering Design* **IJCAI 2026** [\[Paper\]](#)

EDUCATION

- **Technische Universität Kaiserslautern-Landau (RPTU)** Kaiserslautern, Germany
Master of Science in Computer Science *Oct 2023 – Present*
- **National University of Computer and Emerging Sciences (NUCES)** Karachi, Pakistan
Bachelor of Science in Computer Science *Aug 2018 – Jul 2022*

SKILLS

- **Programming Languages**: Python, C/C++, Java, JavaScript, C#, Dart
- **ML Architecture**: VLM-LLM Inference, Transformers, Diffusion Model, Stable Diffusion, LORA
- **Software Engineering Frameworks** : .NET Core, Django, Spring Boot, ExpressJs, NestJs, Flutter, Java Android Development
- **ML Libraries** : PyTorch, PyTorch Lightning, Python OCC, vLLM, Blender Python,
- **Interpersonal** : Creative Thinker, Problem solver, Collaborative Engineer and Researcher, Quick and Adaptive Learner

LANGUAGES

English (C1), Deutsch (A2), Urdu (Native)

HOBBIES

- **Sports**: Playing table tennis and cricket.
- **Travelling and learning**: Visited multiple cities across the United States, Greece, Saudi Arabia, Germany, France, Luxembourg, Singapore, and Pakistan, exploring local culture and history.